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Experimental and numerical study of a photovoltaic/thermal system cooled by metal oxide nanofluids

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ABSTRACT

Photovoltaic modules are impacted by overheating, which degrades their conversion efficiency and shortens their lifespan. This work attempts to study the cooling potential for a hybrid photovoltaic/thermal (PV/T) system by employing CuO and Fe_2O_3 nanofluids with 0.2% and 0.3% volume fractions (φ), circulated in a heat exchanger attached at the rear side of the PV module to lower the temperature of the PV cells and improve its energy and exergy efficiencies. Consequently, a 2D numerical model was analysed given the energy balance across the PV/T system layers, which is evaluated in line with measured experimental data. The results indicate that the successfully synthesised nanomaterials could improve the performance of the nanofluids and increase the heat exchange between PV/T system layers, reducing the PV cells' temperature by 23.49% and 34.58% when cooling by CuO and Fe_2O_3 nanofluids at φ=0.3%. Compared to the uncooled PV module, the electrical efficiency has increased by 9.21% and 10.30%, while thermal efficiency increased by 38.9% and 43.3% under cooling with CuO and Fe_2O_3 nanofluids, respectively. Thereby, the exergy destruction (exergy losses) and entropy generation have decreased by about 26% and 68% when cooling with Fe_2O_3 nanofluid, which improved the exergy efficiency by 13% more than CuO nanofluid.

1. Introduction

The increase in energy demand raises concerns due to increased human activities, fluctuating energy supplies, unstable fossil fuel prices and rising pollution. Renewable energy, especially solar and wind sources, have been introduced as a promising replacement to traditional sources due to their reliable performance and suitability to compensate for an energy supply shortage [1,2]. Solar systems efficiently convert sunlight to clean and sustainable thermal and electrical energy and are a better alternative to traditional ones for environmental safety [3]. The PV module, which consists of silicon cells and converts sunlight into electricity, is the most effective and rapidly developed system type among solar systems. However, the temperature of the PV cells and sunlight intensity significantly affect the PV module conversion efficiency [4]. It has been proven that rising PV cell temperature negatively affects the conversion efficiency and causes performance degradation, especially at high temperatures [5]. Besides, the PV conversion efficiency could be affected sharply when the cell temperature exceeds the mark of 25 °C [6,7]. Increased PV cell temperature decreases the quantity of electrical power generated, impairs its performance, and affects the PV module's lifespan [8]. Extracting excessive heat from the PV module is necessary to increase their efficiency since rising PV cell temperature reduces the terminal voltage and, eventually, the PV performance [9]. Accordingly, lowering the temperature of the PV cells is crucial to increase their performance. The effective way to do so is by removing heat from the PV module and converting it into useful thermal energy by cooling the PV module. The photovoltaic thermal (PV/T) system comprises a heat exchanger (absorbing plates and tubes) placed at the rear of a PV module, can produce electrical and thermal energy

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Nomenclature

A	PV/T area (m^2)
C_p	Specific heat (J/kg K)
D_O	Outer diameter (m)
EP	Electrical power (W)
g	Gravity
HTC	Heat transfer coefficient ($W/m^2/K$)
k	Thermal conductivity (W/m K)
I	Current (A)
I_{sc}	Short circuit current (A)
L	Length (m)
$\dot{m}$	Mass flow rate (kg/m^3)
Nu	Nusselt number
R_a	Rayleigh number
S	Solar radiation (W/m^2)
T	Temperature (°C)
u_a	Wind speed (m/s)
V	Voltage (V)
V_{oc}	Open circuit voltage (V)
W	Width (m)

Subscripts

nf	Nanofluid
NPs	NPs
nm	Nanomaterials
bf	Base fluid

Greek symbols

α	Absorptivity
β	Tilt angle
δ	Thickness (m)
ρ	Density (kg/m^3)
ε	Emissivity
μ	Viscosity (Pa.s)
τ	Transmittance
ν	Kinematic viscosity
ϕ	volume fraction (%)
η_{th}	Thermal efficiency (%)
η_{ele}	Electrical efficiency (%)
$\dot{E}x$	Exergy (W)
$\dot{S}_{gen}$	Entropy generation (W/°C)

simultaneously. Thereby, extracting the heat from PV cells is necessary to improve their efficiency which is achievable by circulating fluids inside tubes placed at the rear of the PV module, transforming the extra heat into thermal energy. [10].

Several studies have been conducted using various methods to improve the PV module's efficacy. Among others, a new cooling design was applied experimentally for a PV/T system, considering different configurations to improve electrical efficiency and maximise thermal efficiency [11]. The first design was made of an absorber copper plate with air as a coolant; the second design employed a copper plate with soldered louvre fins and circulated water inside a serpentine tube. In contrast, the third design was similar to the second one but without circulating cooling water. The study showed a drop in the PV cell's temperature by 31.7% for the second design compared with the first design and 25.4% compared to the third design. Besides, the maximum electrical efficiency achieved was 7.66%, while the thermal efficiency was 89.1% when using the second design. Another study carried out by Menon et al. [12] attempted to compare the performance of an unglazed PV/T system and an uncooled PV module by circulating CuO_2 nanofluid in a serpentine coil together with an absorber sheet. This method lowered the PV/T system temperature to 23.7 °C compared with the system cooled by CuO_2 nanofluid and 15 °C when cooled by water, while the uncooled PV module recorded 68.4 °C. Cooling with water has increased the electrical efficiency by 12.32%, while it increased by 35.67% when cooled by nanofluids compared to the PV module without cooling. Besides, the PV/T thermal efficiency was increased by about 58.77% and 71.17%, respectively. Another investigation on the PV module cooling was verified numerically by Bria [13] by circulating water into aluminium tubes attached to the absorbing plate to control the PV cells overheating and generate thermal energy. Several parameters were considered during the CFD simulation, such as the fluid inlet velocity, inner tube diameter, and inlet and outlet ports. The results concluded that cooling by water has decreased the PV cell's temperature by about 26.8 °C more than the PV module without cooling. Besides, increasing tube diameter and circulated fluid velocity has positively improved the electrical power produced and incremented thermal efficiency.

In recent years, nanotechnology have been exploited to enhance thermal and mechanical performance of many applications, including energy [14], construction materials [15], transportation fuels [16] and many other sectors [17]. In the PV domain, a study employed Cu and Al_2O_3 nanofluids as cooling fluids to lower the PV cell temperature [18]. The nanofluids were circulated in tubes soldered on an absorber plate at the rear of the PV module to optimise their efficiency. The impact of numerous factors, such as concentration of NPs in host fluid, tube diameter and mass flow rates, were analysed to demonstrate the effect of those factors on the PV/T system performance. The study results designated that the overall efficiency of the PV/T system was improved by 10.21% at φ=0.2% of Cu nanofluid, which was better than using Al_2O_3 nanofluid. A similar effect was found when the flow rate of 0.03 kg/s increased, in which the overall efficiency was maximised by 7.4%, with no substantial impact on tube diameter on the efficiencies. Ibrahim et al. [6] have experimentally compared three PV modules' performance under the same operating conditions, in which the first and second modules were cooled by water and Al_2O_3 nanofluid and the third one was left without cooling. The power and efficiency of the PV modules were tested under different mass flow rates and volume fractions of nanomaterials in the host fluid. The outcomes showed that cooling by water and Al_2O_3 nanofluids at 0.05% and 0.07 kg/s flow rate decreased the temperature by about 10 °C and 20 °C, respectively. The nanofluid case showed the highest decrease in the PV/T system temperature by 22.83% and improved the electrical conversion efficiency by 15.5%, while the uncooled PV module recorded the highest temperature, reaching 75.5 °C. In the same trend, an indoor experimental study was conducted using SiC nanofluid circulated into inner grooved tubes with twisted tapes inside. This study has employed a phase change material mixed with 1% SiC NPs on tubes in tightly sealed containers to enhance the PV/T system performance [19]. They examined 0.3% and 0.6% volume fractions at 800 W/m^2 solar radiation and flow rate ranging from 0.008 to 0.058 kg/s to study the effects of these factors on the PV/T system efficiencies. The study concluded that the produced power was increased by 44.5%, and the electrical efficiency increased by 10.8% compared to the uncooled PV module due to improved thermal properties of nanofluids, in which the thermal efficiency increased by 83.3%. Another experimental study was carried out to evaluate the effects of ZrO_2 nanofluid on the PV/T energy and exergy at different volume fractions ranging from 0.015% to 0.0275% and a mass flow rate of 0.8 L/min circulated in serpentine tubes [20]. According to the experimental findings, the cooling by 0.0275% ZrO_2 nanofluid, as opposed to cooling with water, resulted in a 10.2 °C temperature drop for the PV cell and a 24.4% increase in total PV/T system efficiency. Cooling by ZrO_2 at 0.3% helped reduce both exergy destruction and entropy production by 7% and 26%, while the exergy efficiency was increased by

66.8% compared to the uncooled PV module. Anandraj et al. [21] suggested a heat exchanger design consisting of a flat tube (spiral-shaped) with a thermal absorber placed on the rear side of the PV module to remove excess heat by circulating both TiO_2 nanofluid and water in the heat exchanger with ranges of flow rates. The results indicated that using TiO_2 nanofluid and water lowered the PV cell's temperature to 11.2 °C and 5.7 °C, increasing the electrical efficiency by 5.28% and 2.93%, respectively. At a volume fraction of 0.1% of TiO_2 nanofluid and 0.083 kg/s flow rate, the thermal efficiency was increased by 69.2%, while cooling by water recorded 56.45%.

Although the above literature analysis showed many advancements when using nanofluids for PV module cooling, an incomplete scientific vision of the PV/T energy and exergy assessment is still missing in many studies. Besides, many studies have overlooked the uncertainty of data measurement and heat transfer (HT) across the system layers. Therefore, the current study offers an energy and exergy assessment for a PV/T system considering the thermodynamic laws using copper oxide (CuO) and iron oxide (Fe_2O_3) nanofluids at different volume fractions, circulated in a heat exchanger (sheet and tube) placed at the rear of the PV module. The proposed 2D numerical model results were confirmed by MATLAB® and compared with the experimental outcomes. An uncertainty analysis was also carried out to strengthen the accuracy of the measured data and provide robust outcomes for further verification and optimisation studies.

2. Experimental setup and model formulation

2.1. System configuration

The hybrid PV/T system involved a PV module (50 W) and a heat exchanger made from serpentine tubes arranged and soldered on a copper absorbing plate placed at the rear side of the PV module. A thermal grease with high thermal conductivity ($k_{adv} = 1.5 W/m{\cdot}K$) and thickness ($\delta_{adv} = 0.5mm$) was used to avoid heat losses due to the air gap between the back side of the module and the absorbing plate and to ensure perfect contact with the heat exchanger. The heat exchanger was covered by insulation layer type SLENTEX of low thermal conductivity ($k_{ins} = 0.019 W/m{\cdot}K$) and thickness ($\delta_{isu} = 10mm$). Moreover, the rear side of the PV/T system was covered by an aluminium sheet ($\delta_{cov} = 1.5mm$) to minimise heat dissipation as much as possible.

To measure the temperature at various locations on the system, 21 thermocouples (T type) with 0.5 °C precision were employed; four thermocouples were placed on the surface and rear of the PV and PV/T system. Further thermocouples are installed within the nanofluid tank, water tank, the system's input and output, and another thermocouple is used to measure the outside ambient temperature. The coil submerged within the water tank and whose outlet is attached to the nanofluid tank served as the system's outlet. A cooling water tank with a capacity of 35 litres is used to cool the exit nanofluid coming from the outlet by circulating into a coil placed in the water tank to dissipate the temperature of the nanofluid. The flow rate is controlled by a flow sensor (Type SS11.303) set at 1.6 kg/s, placed between the inlet port and pump. A data logger (National Instrument, NI cDAQ-9178) was used to record the measurements, which involved 24 channels for recording the voltages, currents, and a solar sensor type (SS11.303, accuracy 0.1 W/m^2) was used to measure solar radiation. Moreover, a flow rate sensor (YF-S201, accuracy 10%) was used for the flow rate measurement. Using IN SignalExpress 2015 software, a data logger reads data every 30 minutes. The PV module and PV/T systems were fixed at a tilt angle of 14.8° on the top roof-open area of the University of Miskolc. Fig. 1 shows the components of experimental systems with instrumentations.

2.2. Mathematical model

The numerical model was used to analyse the impact of CuO and Fe_2O_3 nanofluids on the performance of the PV/T system at various volume fractions and compare the results to those obtained from the experiments. The numerical model comprises five parts: glass cover, PV cell layer, absorbing copper plate, copper tubes, insulation, and outer aluminium cover (Fig. 2). The numerical model describes the mechanism of heat transfer across the layers of the PV/T system, in which a mathematical equation governing each part of the model was proposed. Several assumptions were considered in the heat balance mathematical equations of each layer of the PV/T system by adopting the mathematical balance equation similar to that used by Rejeb et al. [22]. The properties of the proposed model layers are listed in Table 1. To

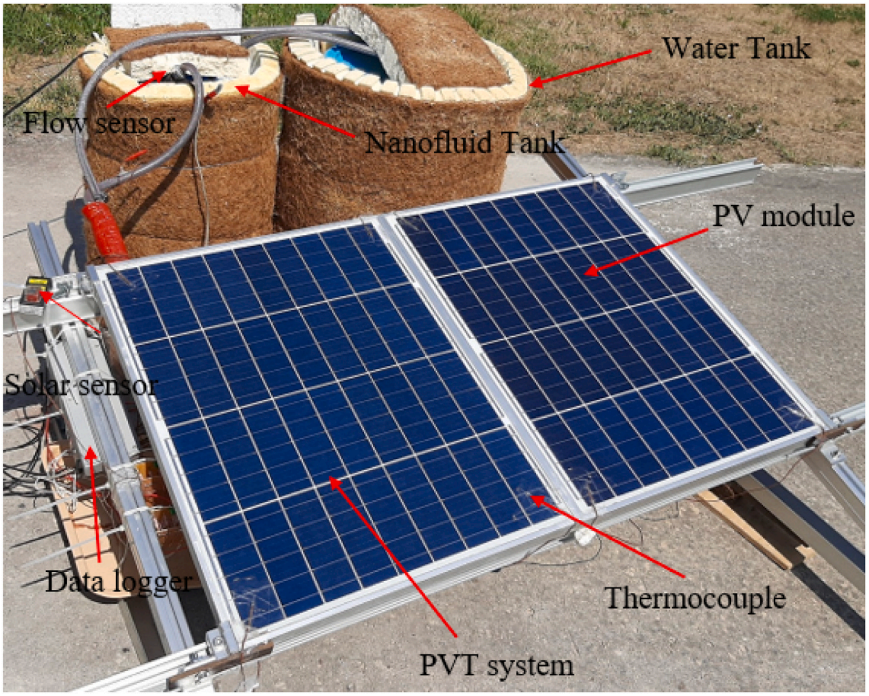


Fig. 1. Experimental set-up.

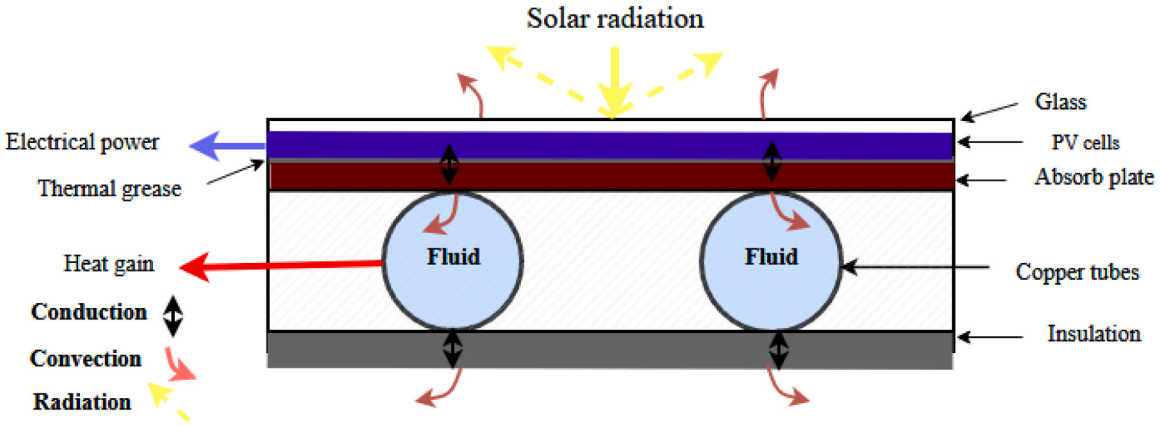


Fig. 2. Sectional scheme of PV/T components.

Table 1
Layers properties of the model proposed.

Glass	PV module	Absorbing plate	Tube
L_{gl}= 0.0025 mm	ρ_{pv} = 2330 kg/m^3	δ_{ab} = 1.5mm	δ_{ab} = 1mm
K_{gl} = layer·K	δ_{pv} = 0.3mm	k_{ab} = 398 W/m·K	k_{tub} = 398 W/m·K
ε_{gl} = 0.88	k_{pv} = 84 W/m·K	ρ_{ab} = 8920 kg/m^3	ρ_{tub} = 8920 kg/m^3
ρ_{gl} = ***2200*** kg/m^3	$C_{p,pv}$ = 700 J/kg K	C_{ab} = 385 J/kg K	$C_{p,tu}$ = 385 J/kg K
δ_{gl} = 3 mm	α_{pv} = 0.95	L×W = 51×48 cm	D_O = 10 mm
C_{pg} = ***480*** J/kg K	ε_{PV} = 0.9		W = 5 cm
α_g = ***0.05***			
τ_g = ***0.936***			

determine the heat transfer coefficient of the PV/T system layers, the formulas in Table 2 were used. The assumptions below were considered while creating the numerical model:

- The solar radiation is uniformly distributed on the outer surface of the PV module, and the thermal conductivity of solid materials is constant.

Table 2
Heat transfer formulas.

HTCs between PV/T system layers	Formula
Radiation HTC between the solar radiation and glass cover of the PV module	$h_{r,gl} = \sigma\varepsilon\left(T_s+T_{gl}\right)\left(T_s^2+T_{gl}^2\right)\left(T_{gl}-T_s\right)$
Convection HTC between the glass cover and PV cells	$h_{pv\text{-}gl} = \frac{1}{\frac{\delta_{pv}}{\lambda_{pv}}+\frac{\delta_g}{\lambda_g}}$
Convection HTC between solar radiation and PV cells	$h_{r,pv} = \frac{\sigma\left(T_{pv}^4-T_{gl}^4\right)}{\frac{1}{\varepsilon_{pv}}+\frac{1}{\varepsilon_{gl}}-1}$
Convection HTC between PV laminate and ambient air	$h_w = 3u_a + 2.8$
Conduction HTC between the PV cells and the absorbing plate	$h_{pv,abs} = \frac{k_{ad}}{\delta_{ad}}$
Conductive HTC between the absorbing plate and tube	$h_{abs,tub} = \frac{2k_{abs}}{(W-D_o)}$
Convection HTC between the tube and nanofluid	$h_{tub,nf} = \frac{Nu_{nf}k_{nf}}{D_{tub}}$
Conductive HTC between the tube and insulation	$h_{\text{tub ,insu}} = \frac{1}{\frac{\delta_{tub}}{k_{tub}}+\frac{\delta_{insu}}{k_{insu}}}$
Conductive HTC between the insulation and outer cover	$h_{insu,bcov} = \frac{1}{\frac{\delta_{insu}}{k_{insu}}+\frac{\delta_{b,cov}}{k_{b,cov}}}$

- The sky is a black body.
- The fluid flowing into the tubes is uniform.
- The layers of Ethylene-vinyl acetate and Tedlar polyvinyl fluoride have been neglected due to slight thickness.
- There is no temperature gradient within the thin model layers.
- The components of nanofluids (fluid and NPs) are in thermal equilibrium and considered as a single-phase fluid.
- Each model layer's temperature distribution is uniform, thermodynamically equilibrium, and steady-state conditions.
- The PV module, absorber plate, copper tubing, and insulation all have constant thermophysical characteristics.
- Kinetic and gravitational energy balances are negligible, and the fluid temperature at the inlet and outlet varies linearly.

2.2.1. *Thermal balance equation of glass cover*

The general thermal representation of HT on the glass cover is indicated in Eq. (1) as follows:

$$\rho_g\delta_{gl}C_{p,gl}\frac{dT_{gl}}{dt} = \alpha_{gl}S + h_{r,gl}\left(T_{sky}-T_{gl}\right) + h_w\left(T_{amb}-T_{gl}\right) + h_{pv,gl}\left(T_{pv}-T_{gl}\right) \quad (1)$$

On the left side of Eq. (1) are the energy changes that occur in the layer of PV glass covers. The right side of Eq. (1) indicates the radiation from the sun received by the PV module, whereas the second part represents the radiation transfer between the PV module's glass cover and the surroundings. The third and fourth items represent the HT by convective conductive between the PV module's cover glass and the environment and among the PV cells and the glass cover.

2.2.2. *Thermal balance equation of the PV module*

Eqs. (2) and (3) represent heat radiation and convection between the PV module, heat exchanger and the environment. The difference between the solar radiation absorbed by the PV module and heat conduction from the rear of the PV module to the absorbing plate represents the thermal balance of the PV module [23].

$$\rho_{pv}\delta_{pv}C_{p,pv}\frac{dT_{pv}}{dt} = \alpha_{pv}\tau_g S - E_{\text{elec}} + h_{pv,gl}\left(T_{gl}-T_{pv}\right) + h_{pv,abs}\left(T_{abs}-T_{pv}\right) \quad (2)$$

$$E_{\text{elec}} = SP\eta_{el} \quad (3)$$

where E, P and η_{el} are the power produced and the electrical efficiency, respectively. $h_{pv,abs}$ is the conductive HTC among the absorber plate and the PV laminate, while T_{abs} is the absorbing plate temperature.

2.2.3. Thermal balance equation of absorbing plate

The thermal balance of the absorber plate is calculated when heat is transferred by conduction from the rear surface of the PV module to the absorbing plate by changing the plate's internal energy. Then, the heat moves between the absorber plate and the tube and between the absorbing plate and the insulation, as mathematically represented in Eq. (4) [23].

$$\rho_{abs}\ \delta_{abs}C_{p,abs}\frac{dT_{abs}}{dt} = \&h_{pv,abs}A_{pv,abs}\left(T_{pv}-T_{abs}\right) - h_{abs,tub}A_{abs,tub}\left(T_{abs}-T_{tub}\right) - h_{abs,insu}A_{abs,insu}\left(T_{pv}-T_{insu}\right) \&} \tag{4}$$

2.2.4. Thermal balance equation of tube

The thermal balance of the tube represents the heat convection and conduction between the tube to nanofluids, the tube to the insulation, and the heat transfer from the absorber plate to the tube, as presented in Eq. (5), as presented in Eq. (5) [23].

$$\rho_{tub}\ \ \delta_{tub}C_{p,tub}\frac{dT_{tub}}{dt} = \&h_{abs,tub}A_{bt}\left(T_{abs}-T_{tub}\right) - A_{tub,f}h_{tub,nf}\left(T_{tub}-T_{f}\right) - A_{tub,insu}h_{tub,insu}\left(T_{tub}-T_{isu}\right) \& \tag{5}$$

2.2.5. Thermal balance equation of fluid

The balance equation of cooling fluid circulating in tubes is written by Eq. (6) as follows:

$$m_{nf}C_{P,nf}\frac{dT_{nf}}{dt} = \dot{m}_{nf}C_{nf}\left(T_{out}-T_{in}\right) + A_{tub,nf}h_{tub,nf}\left(T_{tub}-T_{nf}\right) \tag{6}$$

where T_{in}, T_{out} are the inlet and outlet fluid temperatures, respectively. [24].

2.2.6. Thermal insulation

The energy balance equation of thermal insulation represents the heat conduction between the tubes attached with an absorbing plate to the insulation and on the other side between the absorbing plate and the insulation layer placed on tubes, considering Eq. (7) [23], as follows:

$$\rho_{insu}\ \delta_{insu}C_{p,insu}\ \frac{dT_{i}}{dt} = \&h_{abs,insu}A_{abs,insu}\left(T_{abs}-T_{insu}\right) + A_{tub,insu}h_{tub,insu}\left(T_{tub}-T_{insu}\right) + h_{insu,amb}\ A_{insu,amb}\left(T_{insu}-T_{amb}\right) \& \tag{7}$$

where $h_{ins,amb}$ is the convective HTC among the insulation and the air, and $h_{\mathrm{cov},amb}$ is the convective HTC between the aluminium cover plate and the air temperature.

2.3. Energy and exergy assessment

The thermodynamics laws (1st and 2nd) are applied to demonstrate the quantity and actual work (quality) of energy produced by the system [25]. In this study, the electrical and thermal efficiency produced from the PV/T system is analysed to show the effect of cooling fluids on their efficiency. The electrical efficiency (η_{ele}) of the hybrid PV/T system is the EP produced divided to the area of the system and solar radiation, which is calculated by Eq. (8) [26], as follows:

$$\eta_{ele} = \frac{P_{pv}\times FF}{S\times A} \tag{8}$$

P_{pv} is the power produced by the PV/T system calculated from Eq. (9) [27]. V_{pv}, I_{pv}, A are the current, voltage produced and the area of the system. FF represents the fill factor, which is calculated from Eq. (10) [28].

$$P_{pv} = V_{pv}\times I_{pv} \tag{9}$$

$$\mathrm{FF} = \frac{V_{pv}\times I_{pv}}{V_{oc}\times I_{sc}} \tag{10}$$

Thermal efficiency (η_{th}) represents the thermal performance of the hybrid PV/T system, which mainly depends on the useful thermal energy (Q_u), incident solar radiation and PV/T system area, as indicated in Eqs. (11) and (12) [29].

$$Q_u = \dot{m}\quad C_p(T_{out}-T_{in}) \tag{11}$$

$$\eta_{th} = \frac{Q_u}{S\times A} \tag{12}$$

Accordingly, the sum of electrical efficiency and thermal efficiency represents the overall efficiency of the system, which is calculated in Eq. (12) [30]:

$$\eta_{ov} = \eta_{ele} + \eta_{th} \tag{13}$$

The ratio between convection and conduction HT represents the Nusselt number, which is calculated from Eq. (14) [31].

$$Nu = \left(1 + 1.44\left[1 - \frac{1708(\sin(1.8\beta))^{1.6}}{R_a\quad\cos(\beta)}\right]\left[1 - \frac{1708}{R_a\quad\cos(\beta)}\right] + \left[\left(\frac{Racos.\beta}{5830}\right)^{\frac{1}{3}} - 1\right]\right) \tag{14}$$

where β is the system tilt angle, L_{gl} is the space among the PV cells and glass cover R_a is Rayleigh number, which is calculated from Eq. (15), as follows:

$$R_a = \frac{g\left(T_{pv}-T_{gl}\right)L^3}{\nu\quad\alpha} \tag{15}$$

The thermophysical properties of Fe_2O_3 and CuO nanofluids are calculated depending on the TPP of Fe_2O_3 and CuO nanomaterials are listed in Table 3. Table 4 shows the formulas used for calculating the TPP of nanofluids.

Compared to the uncooled PV module, exergy analysis showed the actual performance of systems. Based on the 2nd law of thermodynamics, The quality of electrical exergy and thermal exergy is assessed. Compared to the surroundings, any temperature change affects the exergy because it increases the system's entropy [36]. Fig. 3 illustrates the exergy flow of the system as a united control volume involving the absorbing plate and serpentine tube attached to the rear surface of the PV module.

Eq. (16) indicates the exergy balance of the PV/T system [37].

$$\sum \dot{E}x_{\mathrm{in}} = \sum \dot{E}x_{\mathrm{out}} + \sum \dot{E}x_{\mathrm{loss}} \Rightarrow \dot{E}x_{\mathrm{solar}} + \dot{E}x_{\mathrm{fluid,\ in}} = \dot{E}x_{\mathrm{electrical}} + \dot{E}x_{\mathrm{fluid,\ out}} + \dot{E}x_{\mathrm{losses}} \tag{16}$$

where $\dot{E}x_{\mathrm{solar}}$ is the solar exergy (input exergy), which can be calculated from Eq. (17) [38]. T_{amb}, T_{sun} are the ambient and the sun temperature (5800 K) [39]. $\dot{E}x_{\mathrm{electrical}}$ is the electrical exergy determined by Eq. (18), which is equivalent to the EP energy [38]. $\dot{E}x_{\mathrm{fluid,\ out}}$, $\dot{E}x_{\mathrm{fluid,\ in}}$, and $\dot{E}x_{\mathrm{loss}}$ are the fluid exergy outlet, inlet and exergy loss,

Table 3
Thermal properties of prepared nanomaterials.

Properties	DI Water [32]	Fe_2O_3 [33]	CuO [34,35]
Density	997 (kg/m^3)	5240(kg/m^3)	6310 (kg/m^3)
Thermal conductivity	0.613 (W/m·K)	20 (W/m·K)	32.9 (W/m·K)
Specific heat	4179 (J/kg·K)	650 (J/kg·K)	551 (J/kg·K)

Table 4
TPP of nanofluids.

Parameter	Formula
Density	$\rho_{nf} = \phi{\cdot}\rho_{nm} + (1-\phi){\cdot}\rho_{bf}$
Thermal conductivity	$\frac{k_{nf}}{k_{bf}} = \frac{k_{nm}+2k_{bf}+2\varnothing(k_{nm}-k_{bf})}{k_{nm}+2k_{bf}-\varnothing(k_{np}-k_{bf})}$
Specific heat	$C_{p,nf} = \frac{\phi{\cdot}(\rho_{nm}{\cdot}C_{p,nm})+(1-\phi){\cdot}(\rho_{bf}C_{p,bf})}{\rho_{nf}}$
Viscosity	$\mu_{nf} = \frac{\mu_{bf}}{(1-\phi)^{2.5}}$
Volume fraction	$\phi = \left[\frac{\frac{m_{nm}}{\rho_{nm}}}{\frac{m_{nm}}{\rho_{nm}}+\frac{m_{bf}}{\rho_{bf}}}\right]\times 100$

respectively.

$$\dot{E}x_{\text{solar}} = S\left(1-\frac{T_{amb}}{T_{sun}}\right) \tag{17}$$

$$\dot{E}x_{\text{ele}} = V_{\text{oc}}.I_{\text{sc}}{\cdot}\text{FF} \tag{18}$$

Eq. (19) used to calculate thermal exergy $\dot{E}x_{\text{th}}$ [38]. Accordingly, the thermal exergy of the system and electrical exergy efficiency are determined by Eqs. (20) and (21) [40].

$$\dot{E}x_{th} = \dot{m}_f{\cdot}C_{p,f}\left[(T_{\text{out}}-T_{\text{in}})-T_{\text{amb}}\ln\left(\frac{T_{\text{out}}}{T_{\text{in}}}\right)\right] \tag{19}$$

$$\eta_{\dot{E}x_{th}} = \frac{m_{nf}{\cdot}C_{p,nf}\left[(T_{\text{out}}-T_{\text{in}})-T_{\text{amb}}\ln\left(\frac{T_{\text{out}}}{T_{\text{in}}}\right)\right]}{S\left(1-\frac{T_{amb}}{T_{\text{sun}}}\right)}\times 100 \tag{21}$$

$$\eta_{\dot{E}x\ \ ele} = \frac{V_{oc}.I_{sc}.FF}{S\left(1-\frac{T_{amb}}{T_{\text{sun}}}\right)}\times 100 \tag{22}$$

The electrical and thermal exergy efficiency of the system represents the exergy efficiency of the PV/T system, which is determined by Eq. (23).

$$\eta_{\ \dot{E}x} = \frac{\dot{E}x_{th}+\dot{E}x_{el}}{\dot{E}x_{solar}}\times 100\% \tag{23}$$

Due to HT losses and the frictional loss in the tubes, which causes exergy destruction (exergy loss), this parameter is determined by Eq. (24) [41] as follows:

$$\dot{E}x_{\text{loss}} = \dot{E}x_{\text{in}} - \dot{E}x_{\text{ele}} - \dot{E}x_{\text{th}} \tag{24}$$

Another thermodynamic parameter is the entropy generation of the PV/T system, which is calculated from Eq. (25) [42].

$$\dot{S}_{gen} = \frac{\dot{E}x_{\text{loss}}}{T_{\text{amb}}} \tag{25}$$

2.4. Synthesis of nanomaterials

2.4.1. Synthesis of copper oxide NPs (CuO NPs)

CuO NPs were prepared using the solvothermal method. The preparation steps are detailed below: copper acetate monohydrate was dissolved in 100 ml of EtOH under vigorous stirring for 30 min. Then, the precursor was completely dissolved and transferred to an autoclave for 12 h at 150 °C. The obtained product was washed and annealed for 2 h at 500 °C. The XRD test was conducted to identify the crystal structure of NPs and can be observed as indicated in Fig. 4-a. Transmission electron microscopy (TEM) was performed to gain more information about nanoparticle morphology. As shown in Fig. 4-b, the NPs have a spherical shape with a diameter range of (80−100) nm.

2.4.2. Synthesis of iron oxide (Fe_2O_3)

Iron oxide NPs were prepared as follows: iron (III) chloride hexahydrate was dissolved in 100 ml of DI water to get a homogeneous solution. Under continuous stirring, NaOH was added drop by drop till the pH solution reached 12. After that, the solution was poured into a stainless-steel autoclave for 9 h at 90 °C. The obtained product was washed several times with DI water to neutral pH, dried and annealed for 2 h at 500 °C. The crystal structure was identified using XRD analysis, as shown in Fig. 5-a. Fig. 5-b shows the TEM, revealing the Fe_2O_3 nanorod (NR) morphology.

2.4.3. Preparation of nanofluid

The two-step method is applied to prepare nanofluids by dispersing the nanomaterials in DI water at different volume fractions, aiming at enhancing the heat transfer rate of the working fluid due to the high TPP of nanomaterials. In the current work, the Fe_2O_3 and CuO nanofluids have been prepared by dispersing 0.2% and then 0.3% in the water. The synthesised nanomaterials were scaled by a scale (BOECO BAS), dispersed in the DI water, and then mixed by stirring for 35 min. Later, a sonication probe (Bransonic, Vf/kHz 240:48) was used for 45 min to achieve good nanomaterial suspension in the DI water for a long period and prevent agglomerate nanomaterials. After finishing the nanofluid preparation, the stability of nanofluids was examined by visualisation and sediment methods by putting a specific amount of nanofluid in a vessel for 4 days without movement after sonication time, as shown in Fig. 6. Then, the required quantities from nanofluids were prepared and used in the experiments. TPP of nanomaterials have a main role in enhancing the HT and the thermal performance of prepared nanofluids.

2.5. Uncertainty analysis of measured data

An uncertainty analysis was performed to specify measurement errors that may occur during the experiment and for more precise measured data. The uncertainty analysis was made for the electrical and thermal efficiencies expressed in Eqs. (26) and (27), considering the method followed by Michael and Iniyan [43]. The uncertainty analysis in the specific fluid heat and the PV/T system area was neglected. The solar radiation voltage, current, flow rate and temperature were

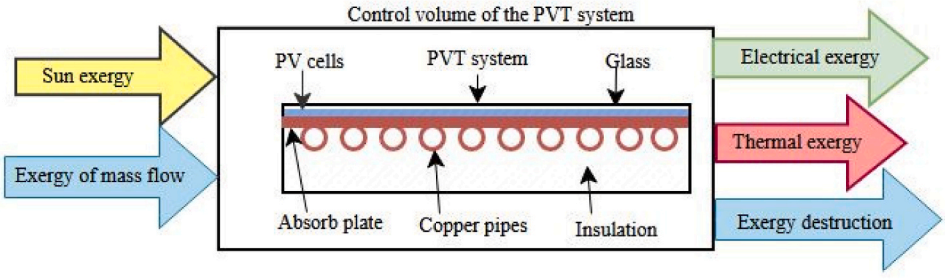


Fig. 3. Exergy flow by a PV/T system.

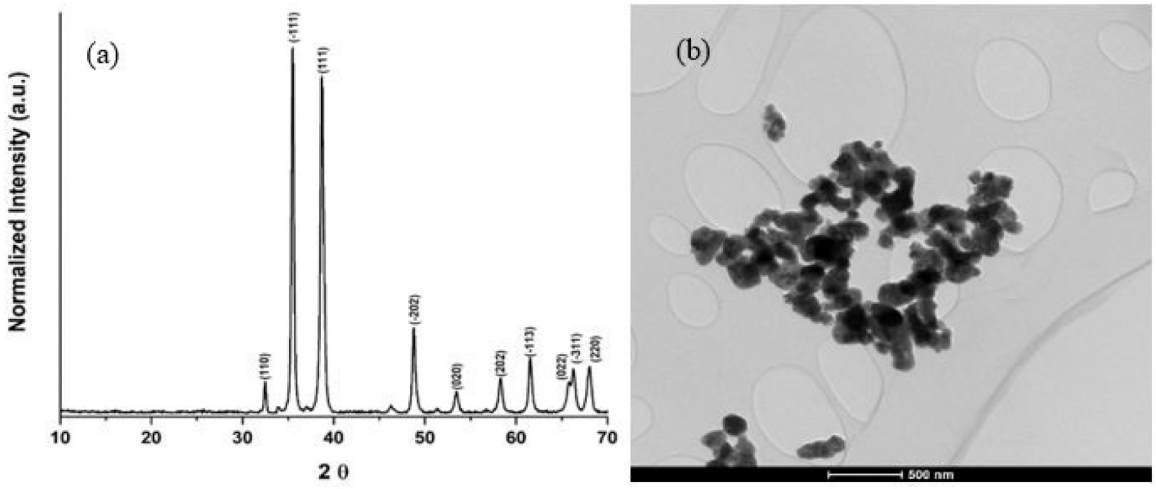


Fig. 4. (a) XRD pattern, (b) TEM image of CuO NPs.

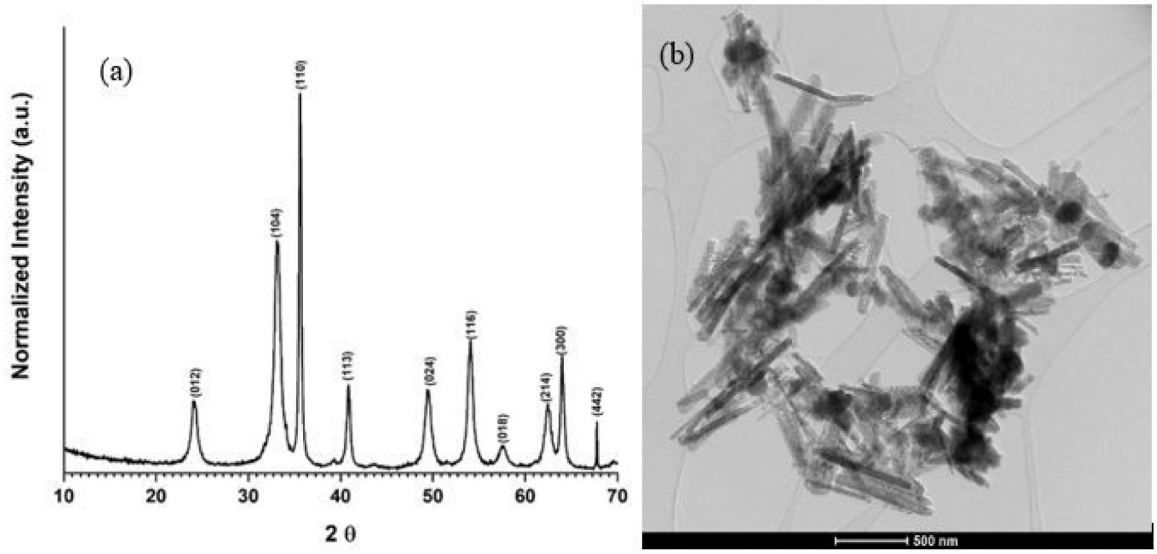


Fig. 5. (a) XRD pattern, (b) TEM image of Fe_2O_3 NRs.

considered to evaluate the uncertainty of measured data, according to Eqs. (28) and (29).

$$\eta_{elec} = f(S, V, I) \tag{26}$$

$$\eta_{therm} = f(S, T, \dot{m}) \tag{27}$$

$$\left[\left(\frac{U_{\eta_{elec}}}{\eta_{elec}}\right)^2\right]_{elec} = \sqrt{\left(\frac{U_S}{S}\right) + \left(\frac{U_v}{V}\right) + \left(\frac{U_I}{I}\right)} \tag{28}$$

$$\& \left[\left(\frac{U_{\eta\ \ therm}}{U_{\eta_{therm}}}\right)^2\right]_{therm} = \sqrt{\left(\frac{U_S}{S}\right) + \left(\frac{U_{\dot{m}}}{\dot{m}}\right) + \left(\frac{U_T}{T}\right)} \tag{29}$$

where U is the uncertainty of the data measured, the uncertainties in electrical and thermal efficiencies were equal to 1.03% and 1.37%, respectively.

3. Results and discussion

The measurements were carried out in Miskolc City, Hungary, to measure the solar radiation and ambient temperature from 8:30–15:00. From Fig. 7, the solar radiation values recorded a maximum of 947.24 W/m^2, with the highest ambient temperature of 34.45 °C, specifically at noon. It can be noticed that the ambient temperature rises with the intensity of solar radiation increases, then eventually reduces with sunset.

3.1. Validation of numerical model with experimental results

The numerical model results were verified by comparing the temperature of the outlet fluid and PV module to the experiment results. The outdoor temperatures were measured from 9:00–15:00 every 30 min of Fe_2O_3 and CuO nanofluid at different volume fractions. Fig. 8 shows the outlet temperature of nanofluids for experimental measurements and numerical model calculations under different volume fractions. A good agreement could be observed between experimental and numerical outcomes with deviations of 1.87% and 2.03% when using Fe_2O_3 nanofluid, 1.72% at volume fraction 0.2% and 1.66% at 0.3% when using CuO nanofluid, compared to 2.05% when using DI water. Fig. 9 shows the PV temperature for experimental measurements and numerical calculations using different nanofluids. The difference between numerical and experimental results of the PV temperature recorded

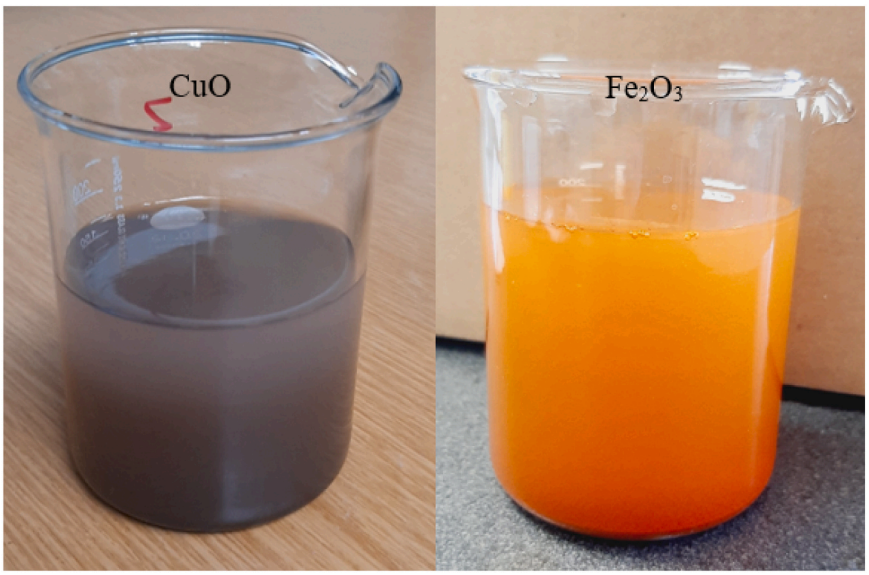


Fig. 6. Stability of nanofluids after 4 days.

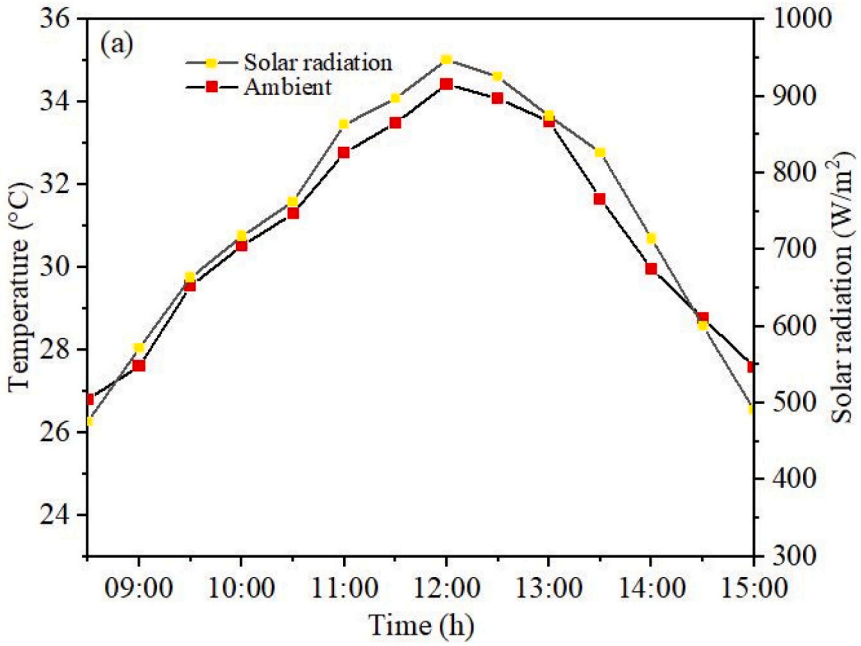


Fig. 7. Ambient temperature and solar radiation of the location under study.

deviations of about 1.85% and 2.21% when using Fe_2O_3 nanofluid, 2.18% and 1.91% when dispersed 0.2% and 0.3% of CuO NPs in DI water, while cooling by water recorded 1.97%. The discrepancies between experimental and numerical results for fluids outlet temperatures and PV temperature variation are attributed to the uncertainties of measurements and assumptions applied for the numerical model. This is common in experimental and numerical studies.

3.2. Energy analysis

3.2.1. PV/T system thermal behaviour

The thermocouple distributions on the surface, the rear side of the photovoltaic thermal system cooled by nanofluids, and DI water indicate the impact of cooling on the temperature reduction of the PV cells throughout the experiment compared with the uncooled PV module. The uncooled PV module recorded a higher temperature than the PV/T, reaching 52.88 °C due to the rising temperature on the surface because of the direct exposure to sunlight. The cooling by nanofluids helps increase heat exchange among the absorbing plate with tubes, which represent (less hot than the rear side of the PV module) and the rear of the PV module, which represents the hottest side, enhancing the heat convection between circulated nanofluids in the wall and conduction of the tube with the absorbing plate, decreasing the PV cells temperature. The thermal properties (thermal conductivity and HT, etc.) of the nanomaterials dispersed in the host fluid (DI water) have increased the heat convective between the tube wall and CuO nanofluid. It led to a lower PV cell temperature depending on the volume fractions of the CuO NP in the host fluid. Dispersion of CuO NPs in DI water at 0.2% lowered the temperature of the PV/T system to 45.32 °C, 0.3% the temperature lowered to 42.82 °C, as shown in Fig. 10. In other meaning, lowering the

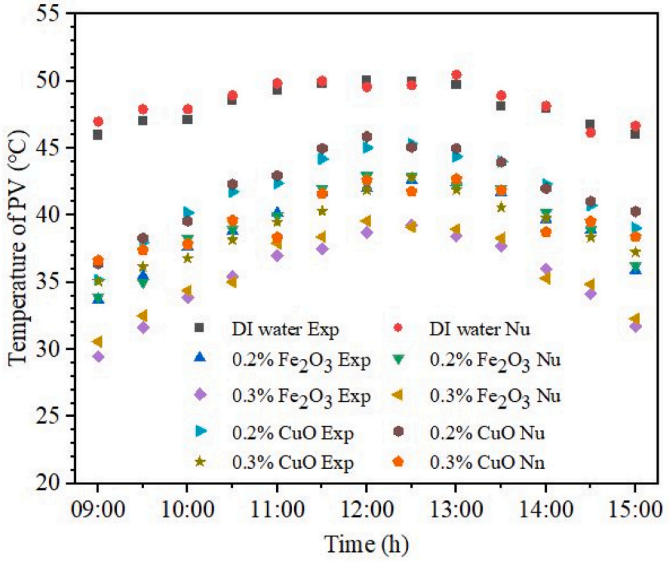


Fig. 8. PV temperature variation when cooled by Fe_2O_3 and CuO nanofluids.

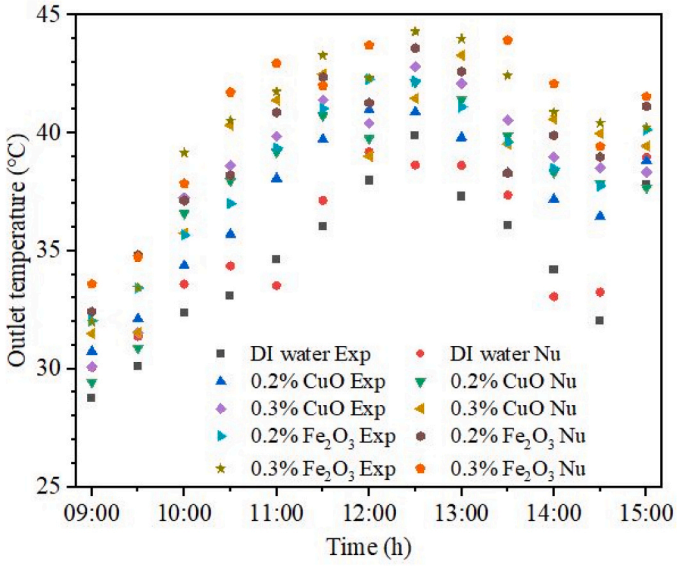


Fig. 9. Outlet temperature variation for PV/T system using Fe_2O_3 and CuO nanofluids.

PV cell's temperature by 16.68% and 23.49% compared to an uncooled unit.

Using Fe_2O_3 nanofluid achieved a high lowering in the PV cell's temperature, more than DI water and CuO nanofluid as a cooling fluid when compared with an uncooled PV module. Dispersed 0.2% of Fe_2O_3 NRs in the DI water lowered the PV cell's temperature to 42.62 °C; at φ = 0.3%, the temperature lowered to 39.31 °C due to improved HT between layers of the PV/T system. Therefore, circulating of Fe_2O_3 nanofluid at (φ = 0.2% and 0.3%) in the heat exchanger lowered the temperature by 24.12% and 34.58%, compared with the uncooled unit, which emphasises the potential of Fe_2O_3 nanofluid as a working fluid compared with DI water, which lowered the temperature to 49.98 °C. This achievement is due to the high TPP of the synthesised nanomaterials and their purity, which improved the system's performance.

3.2.2. *Electrical power*

The EP produced by PV and PV/T systems is measured considering

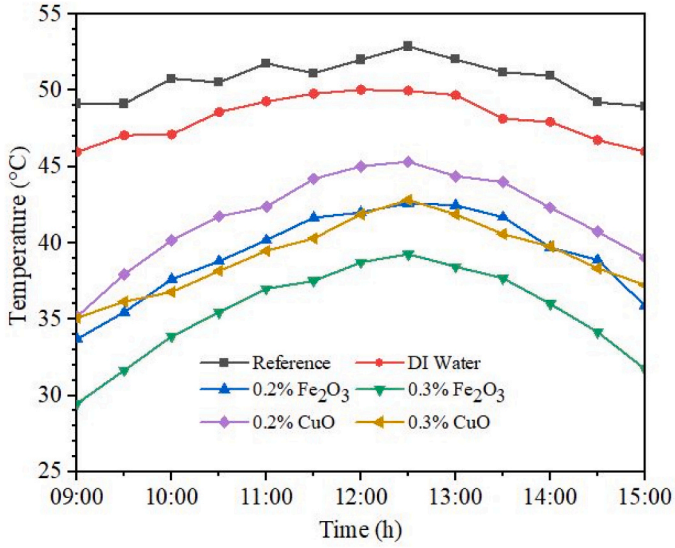


Fig. 10. PV/T system temperature cooled by different nanofluids.

voltage and current, which are impacted by rising the temperature and the intensity of solar radiation on the system's surface. Fig. 11 shows the EP produced by an uncooled PV module and PV/T system cooled with CuO nanofluid. Suspension 0.2% and 0.3% of CuO NPs in water used for cooling have reduced the PV/T temperature and increased the EP to about 30.27 W and 31.74 W, respectively, more than cooling by DI water for cooling, which recorded approximately 21.38 W. Increased HT rate between the PV/T system layers because of the good thermal properties of the CuO NPs dispersed in water has enhanced EP generation because of the effective cooling of nanofluid. Besides, suspended CuO NPs in the water have confirmed the potential of CuO nanofluid to extract rising heat from the PV cells and increase their EP generation by 65.41% and 73.44% more than uncooled the PV module without.

Fig. 11 indicates the improvement in EP produced by the PV/T system using Fe_2O_3 nanofluid at (φ=0.2% and 0.3%). The figure displays a high cooling potential for the Fe_2O_3 nanofluid at both volume fractions, followed by CuO nanofluid and the cooling with water. This is attributed to the role of nanomaterials to effectively increase the heat exchange between the heat exchanger and the rear side of the PV module, improve the HT (convection and conduction), and lower the PV/T system temperature of the uncooled unit. Improvement of the thermal characteristics of nanofluids has increased the heat convection among the nanofluid in tubes and conduction with the rear of the PV/T system, lowering PV cells' temperature and eventually enhancing the EP generated. Considering the best case, an increment could be observed in the PV/T system EP with the increased Fe_2O_3 NR concentration in the base fluid. Accordingly, the EP generation was increased to 33.72 W at suspension 0.2% and 35 W when increased to φ=0.3%, while that of the uncooled unit was about 18.3 W. Consequently, using Fe_2O_3 nanofluid improved the EP by approximately 83.35% and 91.25% over the uncooled PV module.

3.2.3. *Electrical and thermal efficiency*

Decreasing the temperature of the PV cells has improved the EP produced and then enhanced the electrical efficiency and overall efficiency of the system, as shown in Fig. 12. It is worth highlighting that raised temperatures caused the degradation of the PV cell's work and lowered the electrical efficiency of the uncooled PV module to 5.75%. Circulation of CuO nanofluid in heat exchanger reduced temperature and increased the electrical efficiency to 9.21% at φ=0.2%, then increased by 10.30% when φ= 0.3%. Besides, the circulation of nanofluids helps decrease the high temperature from the back surface of the PV/T system and improves its power, thermal energy and efficiency.

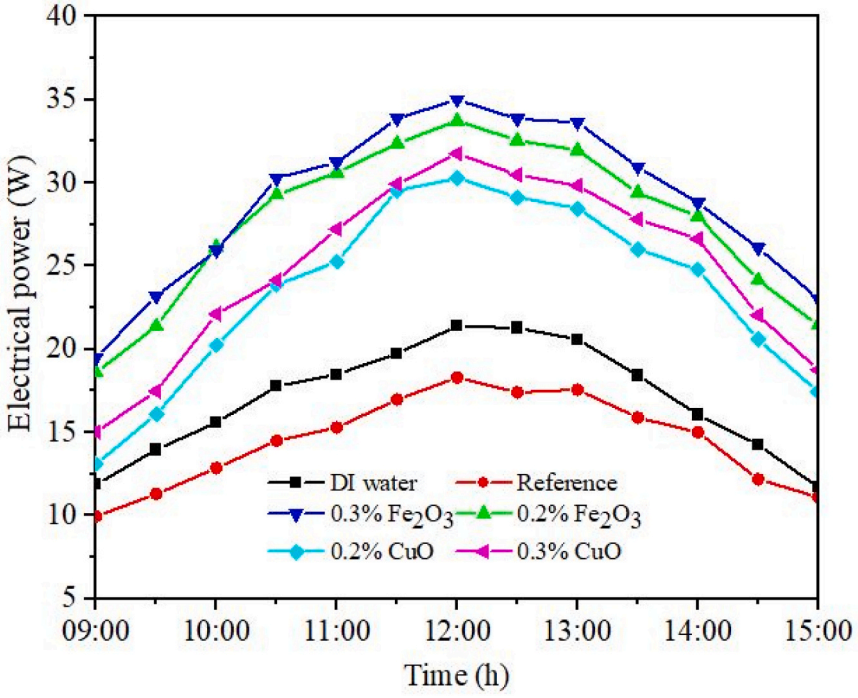


Fig. 11. EP produced of PV/T system under cooling Fe_2O_3 and CuO nanofluids.

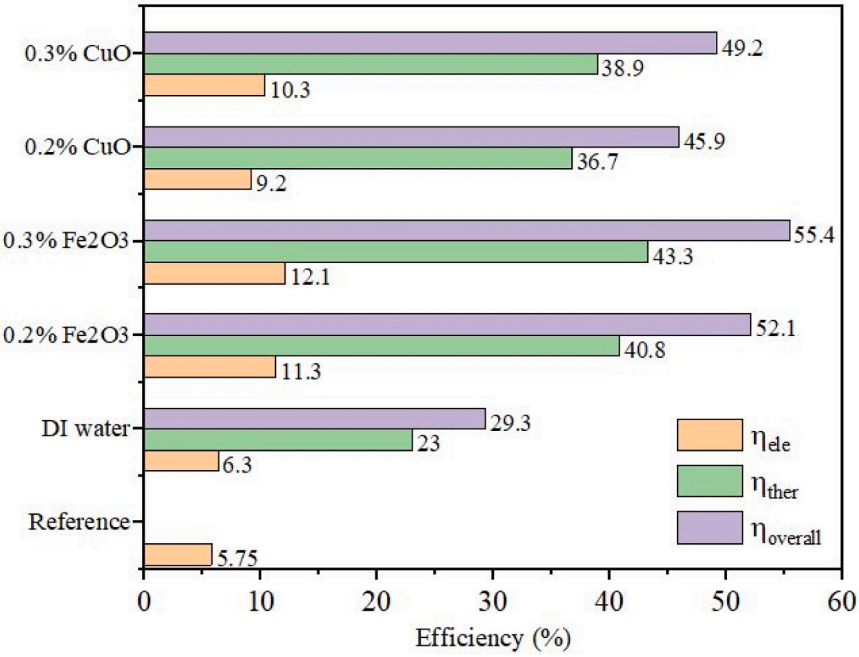


Fig. 12. PV/T system efficiency when cooled with different nanofluids and DI water.

Fig. 12 shows the influence of increased volume fractions of CuO nanoparticles in water on increased PV/T thermal efficiency. It could be realised that the system's thermal efficiency was improved by 36.70% and 38.9%, respectively, compared to 23% attained by using DI water cooling fluid.

Circulating nanofluid in the heat exchanger (tubes integrated with the absorber plate) attached at the rear of the PV module contributes significantly improved heat conduction between the hot area (rear side of the PV module) and the lowest hot area (heat exchanger). Fig. 12 also shows that the cooling with Fe_2O_3 nanofluid has achieved better efficiency because of their high thermal properties compared with water, then led to dropping the temperature and increasing its electrical efficiency by about 11.31% at φ=0.2% and 12.17% at φ=0.3%. This has gradually increased thermal efficiency when the volume fraction of Fe_2O_3 NRs increased in water. The maximum thermal efficiency noticed when using Fe_2O_3 nanofluid is 40.80% at φ=0.2%, which increased to 43.3% at φ=0.3%. Accordingly, the overall efficiency was about 52.10% at φ=0.2% and 55.41% at φ=0.3%

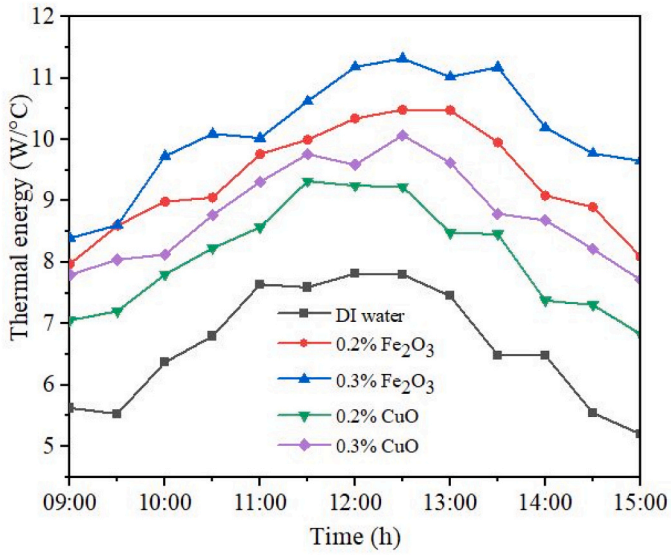


Fig. 13. Effect of volume fraction on PV/T thermal exergy.

3.3. Analysis of exergy results

3.3.1. Thermal exergy

Thermal exergy is the quantity of work that could be obtained from the temperature of outlet fluid and ambient temperatures. Considering Eq. (19), the thermal properties of the fluid used (specific heat), temperature variation among the fluid inlet and outlet, and flow rate influences thermal exergy. Fig. 13 indicates the effect of increased nanomaterials concentration on increased thermal exergy. Dispersed 0.3% of Fe_2O_3 NRs in water has increased thermal exergy by 12.31% more than using CuO at the same volume fraction and over the cooling with DI water by about 45%. It can be observed that using Fe_2O_3 nanofluid has increased the PV/T thermal exergy over the use of CuO nanofluid because of the high thermal conductivity of Fe_2O_3 NR, which positively enhanced the thermal behaviour of Fe_2O_3 nanofluid. The convergence of the temperature of the fluid exit of the PV/T system to ambient temperature is negatively impacted by decreased thermal energy and then reduced in the PV/T thermal exergy quality.

3.3.2. Exergy efficiency

Evaluating the energy quality and specifying the system's irreversibilities is necessary to assess the PV/T system performance, which can be determined by considering the entropy generation, exergy destruction and exergy efficiency dependence on the 2nd law of thermodynamics. The successful method to synthesise nanomaterials has improved the purity and properties of synthesised nanomaterials suspension in water, which increased the thermal performance of nanofluid. This positively increased the heat exchange and heat absorption from the back side of the PV module. As a result, the electrical exergy efficiency and thermal exergy efficiency were raised. Electrical exergy efficiency is associated with EP improvement, affected by the PV cell temperature and the photons absorbed by sunlight. Fig. 14 indicates the influence of increased volume fractions of Fe_2O_3 NRs and CuO NPs in DI water on increased electrical exergy efficiency due to decreased PV cell temperature (as proven in Fig. 10). A remarkable increase in the electrical exergy efficiency with Fe_2O_3 nanofluids was calculated at about 11.18% compared with 9.2% when cooling by CuO nanofluids at 0.3%.

A slight thermal exergy efficiency increment was observed with increased volume fractions in which the maximum thermal exergy reached 1.83% with cooling by Fe_2O_3 nanofluid, compared with 1.71% and 0.9% when cooling with the CuO nanofluid and DI water, respectively. The decreased thermal exergy efficiency is attributed to the lower thermal exergy because the ambient temperature is closer to the outlet temperature, as shown in Fig. 14. However, the high thermophysical characteristics of the Fe_2O_3 nanofluid have improved the exergy efficiency by 13.01% thanks to the increased heat exchange between PV/T system layers. Improved HTCs between the PV/T layers enhanced the heat convection and conduction, reducing the PV/T temperature and dropping the exergy destruction.

3.3.3. Exergy destruction and entropy generation

The PV/T system's irreversibility accelerates the heat losses to the environment; therefore, more exergy loss, particularly with increasing solar radiation at noon, could be observed with an uncooled PV module [42]. Fig. 15 shows the influence of PV/T exergy destruction by cooling

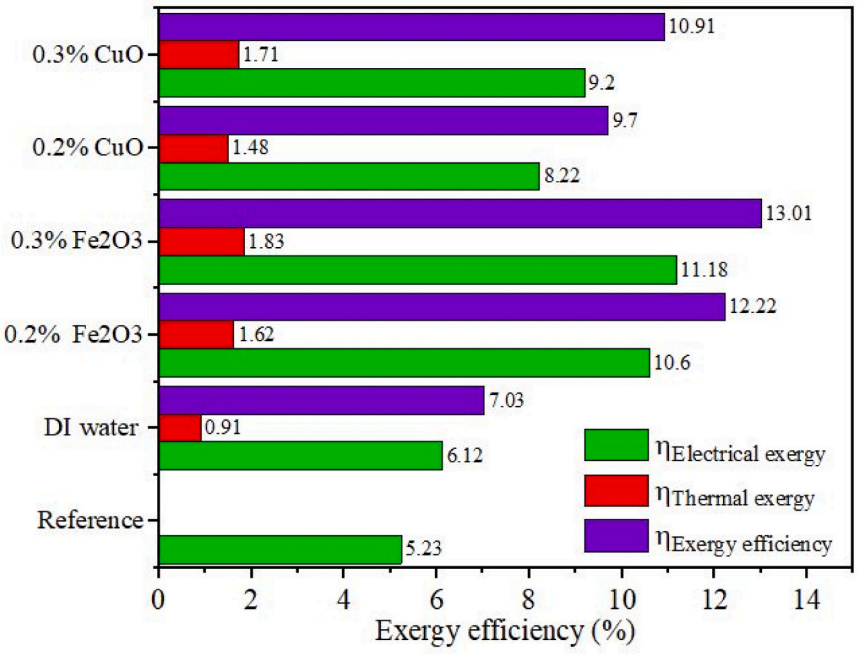


Fig. 14. Effect of increased volume fractions of Fe_2O_3 NRs and CuO NPs on the PV/T exergy efficiency.

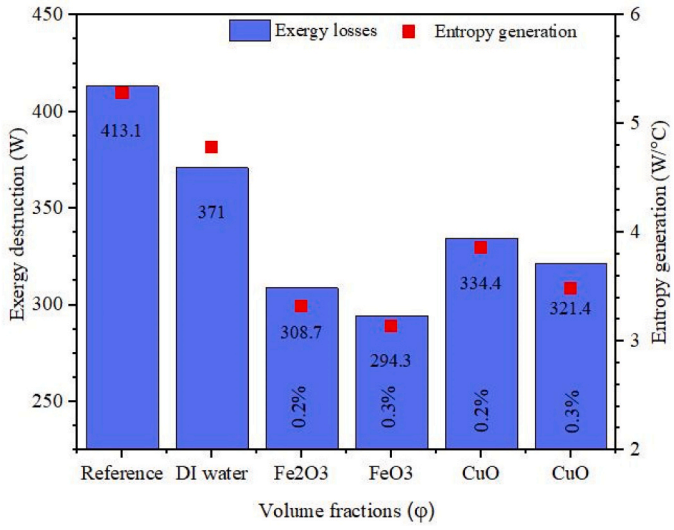


Fig. 15. Effect of cooling by Fe_2O_3 and CuO nanofluids on exergy destruction and entropy generation compared with the DI water.

with nanofluids compared with cooling with DI water. It can be noticed that increased volume fractions of Fe_2O_3 NRs and CuO NPs in the ID water have decreased the temperature and the exergy destruction by 20.18% at φ=0.2% and 26.06% at φ=0.3%, more than cooling by the water. The results showed a slight variation when cooling by CuO nanofluid, in which the exergy destruction increased by 10.95% at φ=0.2% and 15.43% at φ=0.3%. Increasing the temperature between the system and the environment causes an increase in entropy generation, which could be seen in the uncooled PV module, which returns to the rising temperature of the PV cells. Accordingly, lowering the temperature rise of PV cells thanks to nanofluid cooling helps reduce the entropy generation of the system, as indicated in Fig. 15, which illustrates the effect of cooling the PV/T system by nanofluids on the entropy generation. The use of CuO and Fe_2O_3 nanofluid at φ=0.3% has decreased the entropy generation by 51.72% and the entropy generation by 68.15%, which emphasised the potential of Fe_2O_3 nanofluid used as a cooling fluid for solar systems.

4. Conclusion

In this work, the impact of cooling by DI water and nanofluids of different volume fractions on the PV/T thermal and electrical performance has been investigated experimentally and numerically against the uncooled PV module under outdoor weather conditions of Miskolc, Hungary. The key conclusions obtained from the research could be summarised as follows:

- Good agreement was observed between numerical calculations and experimental measurements, with slight deviations due to the uncertainties of measurements and assumptions of the numerical model.
- The successfully synthesised Fe_2O_3 NRs and Cuo NPs have improved the effectiveness of nanofluids, which positively decreased the temperature of PV cells and improved the energy and exergy.
- Cooling by CuO and Fe_2O_3 nanofluids at φ = 0.3% has lowered the PV cells' temperature by 23.49% and 34.58% due to improvement in the heat exchange between the PV/T system layers.
- The electrical power produced has increased thanks to the decreased temperature of PV cells, which improved its electrical efficiency by about 9.21% and 10.30% when cooled by CuO and Fe_2O_3 over the uncooled PV module, which recorded 5.75%.
- Circulating CuO and Fe_2O_3 nanofluids in the heat exchanger has increased the temperature absorption from the rear side of the PV module, incrementing the thermal efficiency by about 38.92% and 43.3% while cooling by DI water enhanced the efficiency by 23% only.
- The PV temperature was reduced as the nanomaterial's suspension (volume fraction) increased in the host fluid, increasing the electrical exergy efficiency by about 9.2% and 11.18% when cooled with CuO and Fe_2O_3 nanofluids, respectively, more than the cooling by water, which achieved 6.12%. The rapprochement of the ambient temperature to the outlet fluid temperature negatively affects the thermal energy, decreasing the thermal exergy efficiency.
- Cooling by nanofluids has influenced the energy losses and entropy generation more than cooling by water. The highest increase in exergy destruction was 26.06%, while entropy generation was reduced by 68.15% when cooling by Fe_2O_3 nanofluid. In contrast, the cooling by CuO nanofluid has increased the exergy destruction and decreased entropy generation by 15.43% and 51.72%.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

Data availability is available on request.

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